

HINTS AND SOLUTIONS

Q.1 B

Sol. $3mg - (T + F) = 3mg$
 $T + mg - F = ma$
 $3mg - T - F = 3T + 3mg - 3F$
 $2F - 4T \Rightarrow T = F/2$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{F}{2\mu}}$$

$$t = \frac{\ell}{v} = \ell \sqrt{\frac{2\mu}{F}}$$



Q.2 C

Q.3 3

Sol. $y(x, t) = 2 \sin^2(k_1x - \omega_1t) \cos^2(k_2x - \omega_2t)$
 $= \frac{1}{2} \{(1 - \cos(2k_1x - 2\omega_1t)) \{1 + \cos(2k_2x - 2\omega_2t)\}\}$
 $= \frac{1}{2} [1 - \cos(2k_1x - 2\omega_1t) - \cos(2k_2x - 2\omega_2t)$
 $+ \frac{1}{2} \{\cos(2k_1x + 2k_2x - 2\omega_1t - 2\omega_2t)\}$
 $+ \frac{1}{2} \{\cos(2k_1x - 2k_2x - 2\omega_1t + 2\omega_2t)\}]$
 $= \frac{1}{4} [1 - 2 \cos(2k_1x - 2\omega_1t) - 2 \cos(2k_2x - 2\omega_2t) + \cos\{2(k_1 + k_2)x - 2(\omega_1 + \omega_2)t\}$
 $+ \cos\{2(k_1 - k_2)x - 2(\omega_1 - \omega_2)t\}]$

Q.4 A

Sol. $y(x, t) = f(x - vt)$
 By comparing
 $v = 0.5 \text{ m/s}$

Q.5 A

Sol. $f_{\text{beat}} = 224 - 218 = 6$
 $\frac{I_{\text{max}}}{I_{\text{min}}} = \left(\frac{4+2}{4-2}\right)^2 = 9$

Q.6 A

Sol. $f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}} = 110 \text{ Hz}$

and $T = \frac{AY}{\ell} \Delta\ell = AY \alpha \Delta\theta$

Q.7 A

Sol. In figure-1, energy at A is potential as well as kinetic. In figure-2 energy at A is only potential.

Q.8 A

Sol. $\lambda f = v = \sqrt{\frac{rRT}{M}}$

$$r = \frac{\lambda^2 f^2 M}{RT} = \frac{100}{g} \times \frac{10^4 \times 50 \times 10^{-3}}{400 \times 25} = \frac{10}{6} = \frac{5}{3}$$

Q.9 A

Sol. $p = \frac{-B \partial S}{\partial x} = -10^{-2} B \times 2p \cos \left(2000 \pi t + \frac{1000 \pi x}{17} \right) \times \frac{50}{17}$

Maximum when

$$\cos \left(\frac{1000 \pi x}{17} \right) = +1$$

Q.10 D

Sol. _{Amit} All collisions are elastic hence velocity will be exchanged.

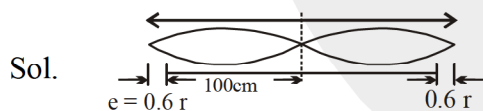
for $0 < t < 4$ $f_1 = f_0 \left[\frac{340}{340 - 40} \right] = f_0 \left[\frac{34}{30} \right]$

for $4 < t < 8$ $f_2 = f_0 \left[\frac{340 - 40}{340} \right] = f_0 \left[\frac{30}{34} \right]$

for $8 < t < 12$ $f_3 = f_0 \left[\frac{340 + 40}{340} \right] = f_0 \left[\frac{38}{34} \right]$

for $12 < t < 16$ $f_4 = f_0 \left[\frac{340}{340 + 40} \right] = f_0 \left[\frac{34}{38} \right]$

Q.11 B

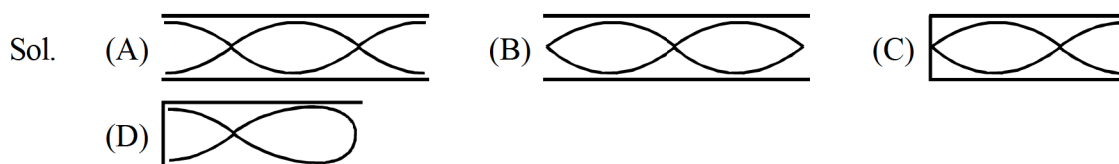


pressure node at $x = -e = -0.6r$ and $x = +100 \text{ cm}$

Q.12 AD

Sol. ABC has negative slope hence, represent compression, while CDE has positive slope hence, represents rarefaction.

Q.13 ABCD



Q.14 BC

Sol. $\frac{v_A}{v_B} = \sqrt{\frac{\gamma_A M_B}{\gamma_B M_A}} = \sqrt{\frac{7 \times 3}{5 \times 5} \times \frac{4}{2}} = \sqrt{\frac{42}{25}}$

(M_A and M_B are molecular weight of A and B respectively)

$$\frac{3v_A}{2l_A} = \frac{3v_B}{4l_B} \Rightarrow \frac{l_A}{l_B} = \frac{2v_A}{v_B} = \sqrt{\frac{168}{25}}$$

Q.15 AC

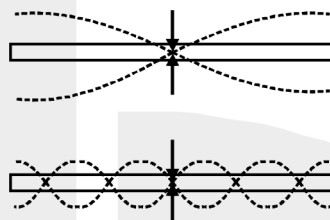
Sol. $L = \frac{\lambda}{2} \Rightarrow \lambda = 2L$

$$\Rightarrow f = \frac{C}{2L}$$

2nd OT $\Rightarrow L = \frac{5\lambda}{2}$

$$\Rightarrow \lambda = \frac{2L}{5}; f = \frac{5C}{2L}$$

f is $\frac{C}{2L}, \frac{3C}{2L}, \frac{5C}{2L}$ etc.



Q.16 BC

Q.17 AB

Sol. $f = \frac{2m-1}{4L} \sqrt{\frac{T}{\mu}} = \frac{2n-1}{4L} \sqrt{\frac{T}{9\mu}}$
 $\Rightarrow \frac{2m-1}{2n-1} = \frac{1}{3} = \frac{3}{9} = \frac{3}{9} = \frac{5}{7}$

Q.18 [Ans. (A) – Q ; (B) – P ; (C) – R ; (D) – S]

Sol._{anu} (A) $\lambda = L$ as $x = 0$ & $x = L$ are nodes $\therefore$ terms should be either $\sin \frac{\pi x}{L}$ or $\sin \frac{2\pi x}{L}$

as $k = \frac{2\pi}{L} = \frac{2\pi}{\lambda} \Rightarrow \lambda = L \therefore$ (Q) is suitable for (A)

(B) $\lambda = 2L$ as $x = 0$ & $x = L$ are nodes $\therefore$ terms should be either $\sin \frac{\pi x}{L}$ or $\sin \frac{2\pi x}{L}$

as $k = \frac{\pi}{L} = \frac{2\pi}{\lambda} \Rightarrow \lambda = 2L \therefore$ (P) is suitable for (B)

(C) $\lambda = 4L$ as $x = 0$ is node $\therefore$ terms should be either $\cos \frac{\pi x}{2L}$ or $\cos \frac{2\pi x}{L}$

as $k = \frac{2\pi}{\lambda} = \frac{\pi}{2L} \Rightarrow \lambda = 4L \therefore$ (R) is suitable for (C)

(D) $\lambda = 2L$ as $x = 0$ is node $\therefore$ terms should be either $\cos \frac{\pi x}{L}$ or $\cos \frac{\pi x}{2L}$

as $k = \frac{\pi}{L} = \frac{2\pi}{\lambda} = \lambda = 2L \therefore$ (S) is suitable for (D)

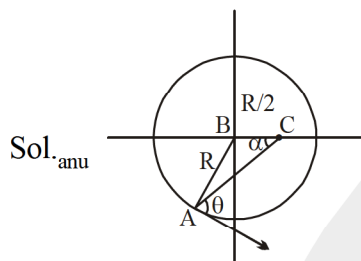
Q.20 [Ans. 0029]

[Sol. $v_1 = \sqrt{\frac{rRT}{m}} = \sqrt{\frac{1.4 \times \frac{25}{3} \times 294}{28 \times 10^{-3} \times 10}} = 350 \text{ m/s}$

$$v_2 = \sqrt{\frac{B}{P}} = \sqrt{\frac{2.1025 \times 10^9}{10^3}} = 1450 \text{ m/s}$$

$$A_t = \frac{2v_2}{v_1 + v_2} A_i = \frac{2 \times 1450}{1450 + 350} \times 18 = \frac{2900}{1800} \times 18 = 29 \text{ Pa}$$

Q.21 [Ans. $6f_0/5$, $6f_0/7$]



For f to be greatest, θ should be minimum in $\triangle ABC$, using sine rule

$$\frac{\sin(90^\circ - \theta)}{R/2} = \frac{\sin \alpha}{R}$$

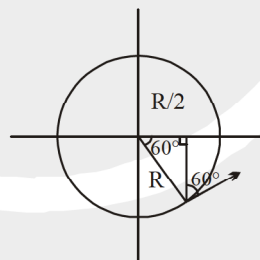
$$\Rightarrow \cos \theta = \frac{\sin \alpha}{2}$$

for θ minimum $\sin \alpha$ should be max. $\Rightarrow \alpha = 90^\circ$

$$\therefore \text{approach velocity} = \left(\frac{C}{3}\right) \cos 60^\circ = \frac{C}{6}$$

$$\therefore f_{\max} = f_0 \left(\frac{C}{C - \frac{C}{6}} \right) = \frac{6f_0}{5}$$

$$\& \quad f_{\min} = f_0 \left(\frac{C}{C + \frac{C}{6}} \right) = \frac{6f_0}{7}$$



Q.22 [Ans. 1.5×10^{-4} J]

Sol. at given time $y = m_1 x$ $0 < x < 0.1$ m
 $= m_2 x + C$ $0.1 \text{ m} < x < 0.15$ m

$$m_1 = \frac{10^{-3}}{0.1} = 10^{-2} \quad \left(V_\omega = \sqrt{\frac{T}{\mu}} \right)$$

$$m_2 = \frac{10^{-3}}{0.05} = 2 \times 10^{-2}$$

$$dK = \frac{1}{2} dm V_P^2 = \frac{1}{2} \mu dx \left(-V_\omega \frac{\partial y}{\partial x} \right)^2$$

$$\int dK = \frac{1}{2} \mu V_\omega^2 \left[m_1^2 \int_0^{0.1} dx + m_2^2 \int_{0.1}^{0.15} dx \right]$$

$$K = \frac{1}{2} T [m_1^2 (0.1) + m_2^2 (0.5)]$$

$$K = \frac{1}{2} T [10^{-4} (0.1) + 4 \times 10^{-4} (0.5)]$$

$$= \frac{10}{2} [10^{-4}] (0.3) = 0.15 \text{ mJ} \quad \text{Ans.}]$$

